

Zhiyuan Zhao

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Overview

Profile

- 8+ years in speech/audio generative systems (ASR, TTS, VC, speech LLMs, and large-scale dataset development).
- 4 publications (2 first-author), 15 patents (7 first-inventor), and 50K+ dataset downloads on Hugging Face.

Interests

- Multimodal speech/audio foundation models for generation, evaluation, and human-AI interaction.
- Controllable generative audio/music systems over synthesis, structure, and effects.

Education

King's College London

M.Sc. in Robotics

2016 - 2017

London, UK · Merit

Central South University

B.Sc. in Electronic Information Science and Technology

2012 - 2016

Changsha, China

Open-Source Projects

Mar 2026 - Present

🔗 **SVLive:** Controllable Music Generation / Live Coding / Synthesizer Programming

- Browser-based live coding music playground (*Strudel* + *Vital* + *LLM*), supporting real-time editable synthesis.
- Support for 2,300+ synthesizer presets with lazy rendering and programmable sound selection.
- Exploration of natural-language control over synth parameters, music structure, and audio-feedback alignment.

Nov 2025 - Jan 2026

🔗 **LEMAS:** Large-scale Multilingual Speech Corpus & Benchmark Models

- 150,000+ hour multilingual speech corpus with word-level timestamps across 10 languages, built via a scalable high-quality data pipeline.
- Benchmark models for speech synthesis (*LEMAS-TTS*) and speech editing (*LEMAS-Edit*) trained on the proposed dataset.

Sep 2025 - Oct 2025

🔗 **VoicePlatform:** Enhancement / Recognition / Synthesis / Editing / Translation

- Multimodal AI platform integrating six capabilities: audio processing(denoise, upsampling and separation), speech recognition, synthesis, editing, video editing, and translation.
- Simple and clear interface for exploring and controlling multiple audio/video AI modules in a unified system.

Apr 2024 - Feb 2025

🔗 **OpenVideo:** High-quality Text-to-Video Dataset and generation framework

- 672+ hour / 106K+ clip / 720p text-to-video dataset sourced from *Pexels-Raw* for generative video research.
- Dataset tooling for annotation, validation, and migration to ensure quality control and consistent video-text alignment.

Industry Experience

2026.03 - Present · XPENG Robotics

Senior Research Engineer

- Driving **multimodal humanoid-robot interaction** with expertise in speech understanding, dialogue robustness, and instruction-following.
- Leading speech data pipelines for iterative improvement of dialogue and language capabilities in human-robot systems.

2022.03 - 2026.03 · International Digital Economy Academy (IDEA Research)

Speech Technology Lead

- Led multimodal speech model development, enabling zero-shot voice cloning and real-time conversational AI for digital humans and agents.
- Built end-to-end AI video translation systems integrating ASR, translation, TTS, and subtitles.
- Deployed speech generation models powering **consumer products with millions of users**.
- Developed multilingual speech systems, including Vietnamese ASR **outperforming Azure and other commercial solutions**.

2019.01 - 2022.02 · UBTECH Robotics Corp.

Senior Speech Synthesis Algorithm Engineer

- Researched and deployed neural speech generation models for voice cloning, expressive synthesis, and cross-lingual speech transfer.
- Optimized **low-latency speech synthesis** systems for edge devices and real-world applications.
- Contributed speech technologies showcased in the **CCTV 2022 Spring Festival Gala**.

2018.04 - 2019.05 · SF Technology Co., Ltd.

Optimization Algorithm Engineer

- Designed and deployed big-data ML pipelines processing **100M+ parcels per day** for forecasting, route planning, and dynamic optimization.
- Automated vehicle reallocation for peak periods, generating **thousands of daily plans** and reducing manual workload by ~75%.

2017.09 - 2017.12 · Samsung Telecom R&D Center

Software Development Intern

- Worked on **Bixby** voice assistant testing, NLP analysis, and multilingual data annotation.

Publications

[1] **LEMAS: A 150K-Hour Large-scale Extensible Multilingual Audio Suite with Generative Speech Models**

Zhiyuan Zhao, Lijian Lin, Ye Zhu, Kai Xie, Yunfei Liu, Yu Li

arXiv:2601.04233, 2026  PDF

[2] **MelTok: 2D Tokenization for Single-Codebook Audio Compression**

Jingyi Li*, Zhiyuan Zhao*, Zhisheng Zhang, Yunfei Liu, Lijian Lin, Ye Zhu, Jiahao Wu, Qiuqiang Kong, Yu Li

arXiv:2510.01903, 2025  PDF

[3] **STFTCodec: High-Fidelity Audio Compression through Time-Frequency Domain Representation**

Tao Feng*, Zhiyuan Zhao*, Yifan Xie, Yuqi Ye, Xiangyang Luo, Xun Guan, Yu Li

IEEE ICME, 2025  PDF

[4] **Improving Model Stability and Training Efficiency in Fast Speed High Quality Expressive Voice Conversion System**

Zhiyuan Zhao, Jingjun Liang, Zehong Zheng, Linhuang Yan, Zhiyong Yang, Wan Ding, Dongyan Huang

ACM ICMI, 2021  PDF

* equal contribution.

Patents

First inventor:

[1] Audio editing method, system, terminal, and storage medium — CN202510616027.1

[2] Training method and synthesis method for a speech synthesis model — CN202410346345.6

[3] Data quality inspection method, device, equipment, and storage medium — CN202310010360.9

[4] Cross-lingual audio conversion method, computer equipment, and storage medium — CN202011516681.9

[5] Cross-lingual speech conversion method, computer equipment, and storage medium — CN202011588869.4

[6] Training method of a speech synthesis model, electronic equipment, and medium — CN202011448005.2

[7] Voice conversion method, device, equipment, and storage medium — CN201980003287.4

Co-inventor:

[8] Interactive digital-actor video generation method, system, device, and computer storage medium — CN202510603049.4

[9] Speech synthesis method, device, terminal equipment, and storage medium — CN202111630461.3

[10] Speech emotion recognition method, terminal device, and computer-readable storage medium — CN202111615879.7

[11] Model training method, device, terminal equipment, and computer-readable storage medium — CN202111682661.3

[12] Adaptive sample dataset generation method, device, and equipment — CN202111632906.1

[13] Speech denoising method, device, electronic equipment, and computer-readable storage medium — CN202111682647.3

[14] Encryption method, device, terminal equipment, and computer-readable storage medium — CN202211152830.7

[15] Voice conversion method, system, device, and storage medium — CN201980003189.0